

Practical Number - 04.

Practical Name = To draw the Structural view dig, class dig, Object dig.

Outcome = class dig helps to understand the static structure of a system. It shows relationships between classes, objects, attributes, and operations.

Objective = Identify the classes. Classify them as weak and strong classes & draw the class dig.

Theory

A class dig models the static structure of a system. It shows relationship between classes, objects, attributes and operations.

Basic class diagram Symbols and Notations.
Classes.

Classes represent an abstraction of entities with common characteristics. Association represent the relationship between classes.

Illustrate classes with rectangles divided into compartments. place the name of the class in the first partition (centered, bolded and capitalized), list the attributes in the second partition (left-aligned, not bolded, and lower case) and write operations into the third.

class Name
attributes
Operations()
responsibility

class.

Active classes initiate and control the flow of activity, while passive classes store data & serve other classes. Illustrate active classes with a thicker border.

Class Name	Active class
attributes	
operations()	

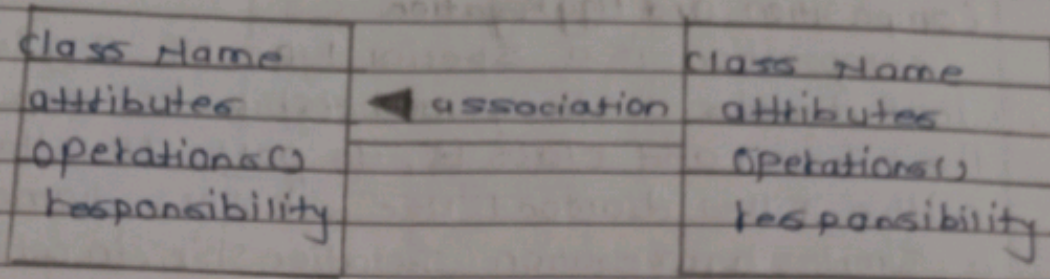
Visibility -

Use visibility markers to signify who can access the information contained within a class. Private visibility, denoted with a - sign, hides information from anything outside the class partitions. Public visibility, denoted with a + sign, allows all other classes to view the marked information.

Class Name	visibility	Marker	visibility
attributes		+	Public
+ Public operation		-	Private
- Private operation		#	Protected
# Protected operation		-	Package

Associations

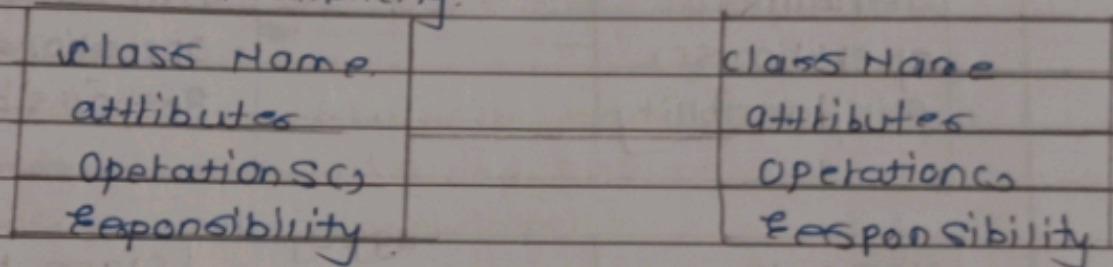
Associations represent static relationships between classes. Place association names above, on or below the association line. Use a filled arrow to indicate the direction of the relationship. Place roles near the end of an association. Roles represent the way.



Multiplicity (Cardinality)

place multiplicity Notations near the ends of an association. These symbols indicate the number of instances of one class linked to one instance of the other class. For ex. One company

Multiplicity.



Indicator

Meaning

0..1

Zero or one

1

one only

0..*

0 or more

1..*|*

1 or more

n

Only n (where $n > 1$)

0..n

Zero to n (where $n > 1$)

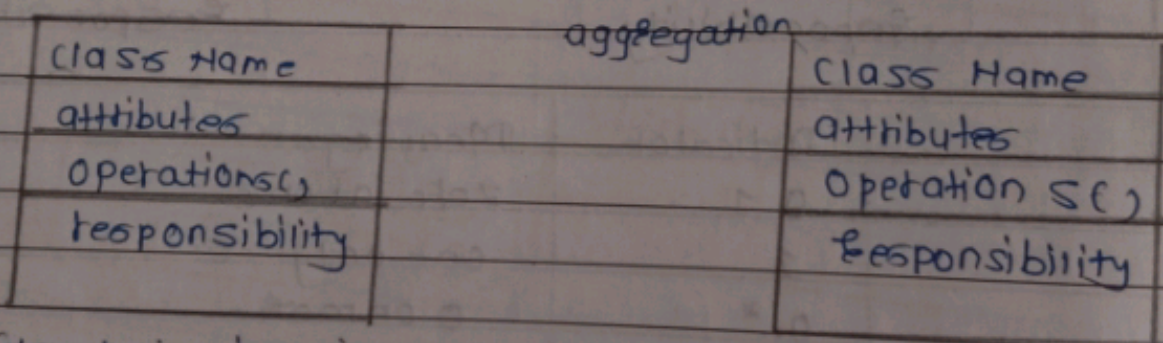
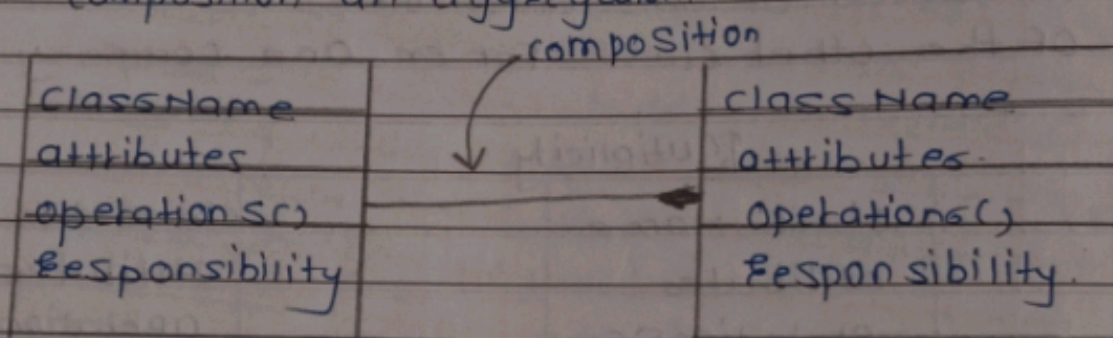
1..n

One to n (where $n > 1$)

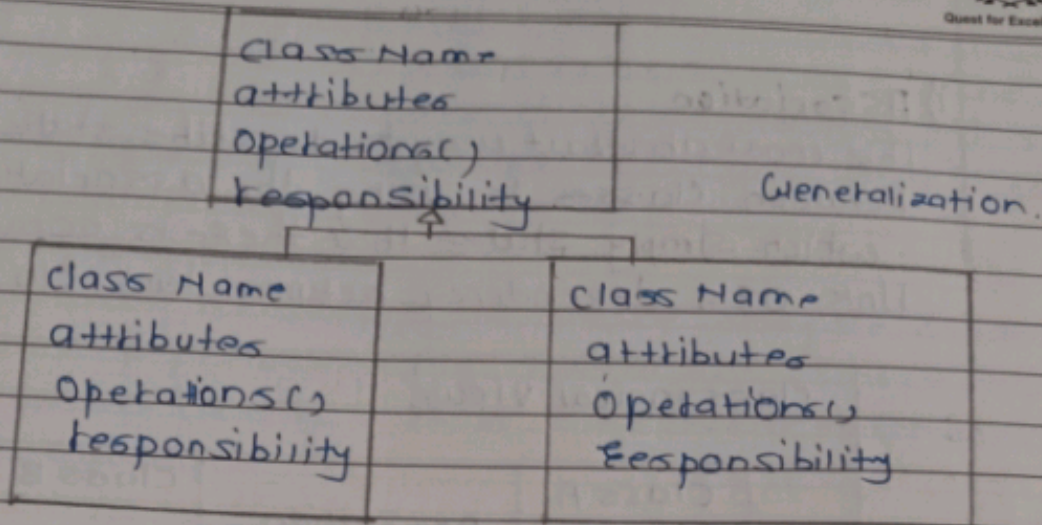
Composition and Aggregation.

composition is a special type of aggregation that denotes a strong ownership between class A, the whole, and class B, its part. Illustrate composition with a filled diamond. Use a hollow diamond to represent a simple aggregation relationship, in which the "whole" class plays a more important role than the "part" class, but the two classes are not dependent on each other.

Composition and aggregation



Generalization is another name for inheritance or an "is a" relationship. It refers to a relationship between two classes where one class is a specialised version of another. For ex.: Honda is a type of car, so the class Honda would have a generalization relationship with the class car.



In real life coding examples, the difference between inheritance and aggregation can be confusing. If you have an aggregation relationship, the aggregate (the whole) can access only the public function of the part class.

On the other hand, inheritance allows the inheriting class to access both the public and protected functions of the superclass.

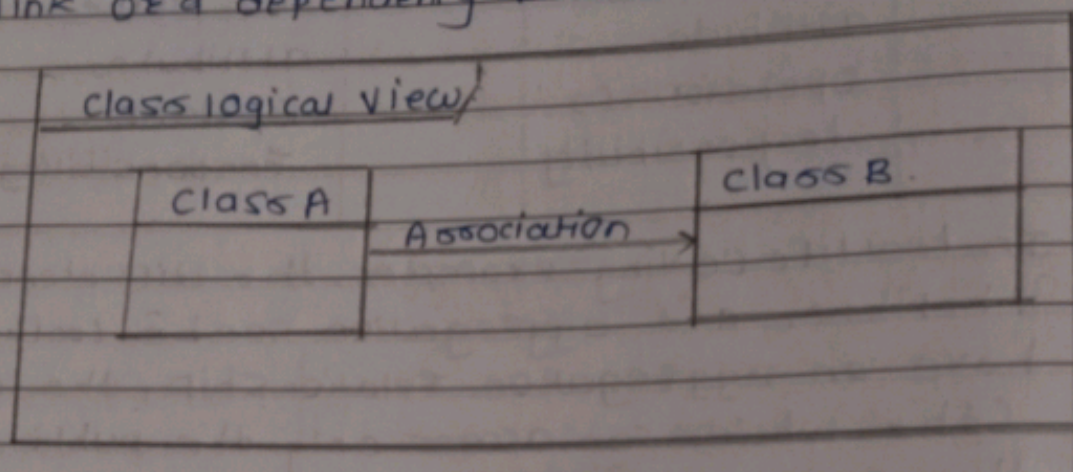
UML class diagram: Associations, Aggregation and Composition

The UML class diagram is used to visually describe the problem domain in terms of types of objects (classes) related to each other in different ways.

- There are 3 primary inter-object relationships: Association, Aggregation and Composition. Using the right relationship is important for placing implicit restrictions on the visibility and propagation of changes to the

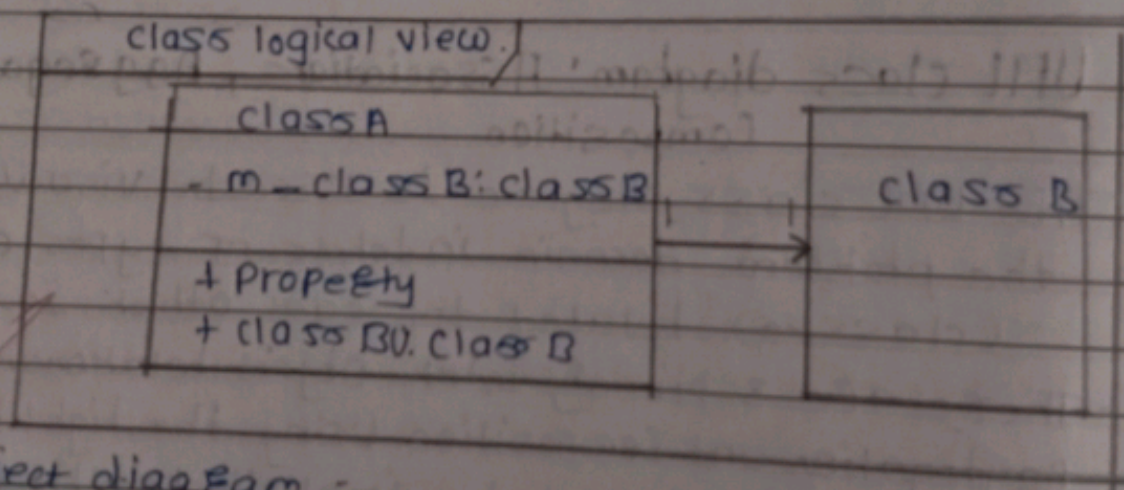
i) Association

- The most abstract way to describe static relationship between classes is using the association link, which simply states that there is some kind of a link or a dependency between two classes or more.



ii) Strong association

Class A may also be linked to class B in order to show that it holds a reference to class B instance.



Object diagram

A UML Object diagram represents a specific instance of a class diagram at a certain moment in time when represented visually, you'll see many similarities to the class diagram.

Object diag are not limited to banking use cases, however as you can easily make an object diagram for family trees, corporate departments or any other system with interrelated parts.

Object diag elements.

Object diag are simple to create: they're made from objects represented by rectangles, linked together with lines. Take a look at the major elements of an object diagram.

Objects.

Objects are instances of a class. For example, if "cat" is a class, a 2007 Nissan Altima

Conclusion - The class diagram & object diag was made successfully by following the steps described above

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